

Computational HDR Photography Algorithm Exploration

Team 13

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Abstract

HDR photography is a method to produce a photo of a high contrast scene with detail in both the lightest and darkest parts of the image. Most digital screens can only display colors with 8 bits per color channel. This leads to problems where there are not enough colors to properly display details in scenes with high variation between the lightest and darkest colors. HDR photos attempt to bring detail back, but there is always a loss of information somewhere because of the hardware limitations. Currently, 2 methods are commonly used to create HDR images: manual creation using photo editing software, and digitally on a camera, often using proprietary software. HDR creation using photo software is more of an art than a science, and therefore will not be discussed further. Camera software does a remarkable job at creating HDR images, but it is often done within a black box with very little information on the exact process. This project reveals the pros and cons of several algorithms to compute one pixel at a time. The two main algorithms used in this project are a simple average algorithm, and a pixel select algorithm based on the “most colorful” pixel (closest to middle gray). Both of these algorithms have obvious problems, and the other algorithm is often the fix for the problem. Because of this, an optimal single pixel HDR algorithm will use a combination of pixel average and pixel selection to find an optimal middle ground.

Keywords

HDR, Dynamic Range, Photography, Computational Photography, Tone Mapping, Color, Exposure

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Introduction and Background

HDR photography is a method to produce a photo of a high contrast scene with detail in both the lightest and darkest parts of the image. Cameras, both digital and analog, have issues displaying scenes with a large difference between the lightest and darkest section of the image, also known as high dynamic range. This is partially an issue with the way that sensors detect the magnitude of the light, but it is mostly an issue of arbitrary data restrictions. Many cameras can record higher resolution color data, but it is only while reading the raw sensor data. Most digital screens can only display colors with 8 bits per color channel, so many popular photo file formats are limited to 8 bits per channel. This leads to problems where there are not enough colors to properly display details in scenes with high dynamic range. HDR photos attempt to bring detail back, but always at a loss of information somewhere else. Bringing details back to a photo adds data, so data needs to be lost somewhere else in order to still fit within the confines of the 8 bit color channels. The lost data loss comes from altering or misrepresenting the colors in the image, which can lead to the image losing aesthetic value.

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The goal of this project is to implement several algorithms to compute an HDR photo one pixel at a time, so the algorithm can be directly compared to the produced result. The two main algorithms used in this project are a simple average algorithm, and a pixel select algorithm based on the “most colorful” pixel (closest to middle gray). Both of these algorithms have obvious problems, and the other algorithm is often the fix for the problem. Because of this, an optimal single pixel HDR algorithm will use a combination of pixel average and pixel selection to find an optimal middle ground.

The next steps in the project involve creating more complex algorithms to produce better photos, aesthetically. There are three obvious areas for improvement: algorithm complexity within the single pixel algorithms, algorithms that compute more than one pixel at a time, and using the raw camera data for computation to take advantage of the higher number of bits per channel. Each path has significant room for aesthetic improvement of the final image, and hopefully will be helped along by the work completed in this project.

Design, Development, and Testing

Design

The design of this project is relatively simple, since the primary goal is to produce images, with minimal thought for any sort of UI. Any time spent developing a UI was time not spent working on the meat of the project. The project is just a collection of C files and headers, two folders for input and output images, and a make file to streamline compilation and running.

The design of the code follows a pretty simple flow pattern. The jpeg images in the source file are converted to char arrays containing the pixel color values[1]. Then the code uses the selected algorithm (set in main.c) to produce a jpeg image in the image output folder.

Using the code is slightly more complex. Source images are taken in rapid succession with minimal camera movement, and each photo is taken at a different ISO. The ISO is adjusted by as few stops as possible between images, while still capturing all desired detail. ISO is used to adjust the exposure to keep the depth of field and motion blur consistent across the source images. The source images go into a folder with some sort of descriptive name. The code is currently set up to process three images at a time. To run the code, file path names need to be updated to make sure the photos read and write to the correct locations, and the HDR algorithm is selected by changing the selector value in main.c.

Development

This project has only one required feature, and so producing HDR photos drove design and development. The basic design is pretty simple: convert input images to some type of array representation, and then produce an image by looping through the arrays to achieve a result.

Development first began on converting RAW camera data to a pixel array format, which was met with minimal success. Development started with working on the raw data because the greater number of bits per color channel make it a superior source of data to any jpeg image.

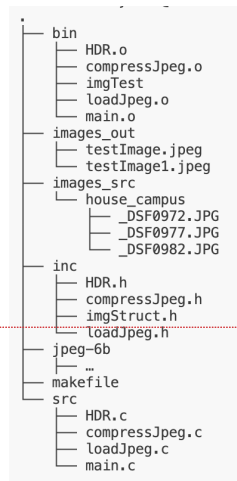


Figure 1 – Project file structure with shortened jpeg-6 file. The source images go in a descriptive folder within images_src.

Commented [LN1]: Add a flowchart?

Commented [LN2]: Add source info?

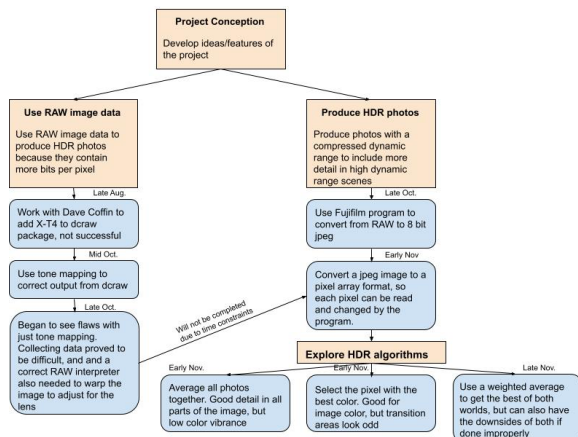


Figure 2 - Development Flowchart showing the path of feature development from conception through completion

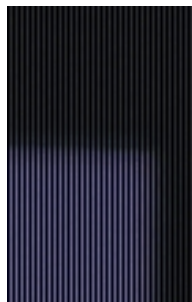


Figure 5 - Zoom in on first image produced. Full photo is excluded for brevity, especially as the lines do not show up even when the image is the full width of the page.

Testing

Testing for this project is easy to do by hand, especially for algorithms which only work on a single pixel at a time. Most of the time, it



Figure 3 -- First pixel select image. Works well in the sky, but somewhat obvious that there is a problem in the leaves with all the rainbowning

Turning the raw data into a reasonably accurate pixel array proved to be challenging, and got set aside for using jpeg source images instead. Getting life-accurate colors and image warping is a project all on its own, with its own challenges and victories to be found.

The source photos for this project were taken using a Fujifilm X-T4, which produces both raw and jpeg files. Jpeg is a

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well known and commonly used format, and it is relatively easy to convert into a pixel array with a small amount of code. To test if the conversions between jpeg and a pixel array were working properly, the first image of the project was produced by turning every other pixel in an image to black (Figure 3). Once the pixel arrays work, it is a simple leap to implementing simple algorithms like the simple average.

Figure 4 - Failure from a more complex algorithm which chooses the 2 best photos and takes a weighted average based on how "colorful" each pixel is.

will be obvious when something goes wrong, as seen in Figures 4 and 5, but it is also trivial to compare the expected result for a pixel to the actual result, based on the values of the input pixels.

Results

Overall, the HDR project was a success, although it lacks some of the complexity that was originally planned. The project produces HDR images using a variety of algorithms, and sets a good foundation for further exploration and learning. However, some of the failures, particularly in relation to processing raw camera data, meant that the project is currently restricted to 8 bit input data and there was not enough time to explore more complex algorithms before the end of the semester.

At this point, the project contains the following features: it accepts jpeg image data, and produces a jpeg image with higher detail using one of the selected algorithms.

The biggest problem occurred when working on processing the raw camera data. Because of the potential benefit of higher bit source data, too much time was spent on this part of the project. Information on raw data is difficult to come by because it is often guarded by the camera manufacturer, but the theory is simple so development always felt one small step away from a breakthrough. This led to a stagnation in progress, which negatively affected the schedule of the rest of the project.

After this large issue with scheduling and estimates, very little problems occurred because of the experience gained. Although it was a large problem, the rest of the project flowed smoothly by comparison, especially because of the skills learned. Most of the time spent on this project was focused on learning new concepts and skills. The project would be trivial to repeat within a week, now that the team has all of the knowledge and experience of going through it once already.

Finally, the results:



Figure 6 – 3 example input images, taken on a Fujifilm X-T4



Figure 7 - Simple Average image



Figure 8 – Simple Select image

Conclusions and Future Work

Photos suffer from both under and over exposure, which is quite different from the way humans experience seeing the world. HDR photos attempt to counteract this problem by taking many images at different exposures, and using all of them to fill in the details for the entire image. This process is quite common, but usually done manually or by the camera before the user gets to even see the image. This project is the first step in algorithmically producing HDR images while allowing the whole process to be observed, from start to finish. Every aspect of this project is done better by large photo companies, but their software leaves very little room for learning. Going forward, this project can be expanded to use raw camera data, as well as use more complex algorithms to produce images that become true works of art.

References

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- D. Coffin, dcraw. RAW camera data processing package